Session 8: GNSS Constellation Dynamics and Navigation

Task 1: Explain why four satellites are needed for a fix, and what dilution of precision means geometrically

Why are four satellites are needed for a fix:

- **3 Spatial Unknowns** (x, y, z): To determine a position in 3D space, the receiver must solve for its three Cartesian coordinates.
- **1 Clock Bias Unknown** (b_0): Standard receivers use inexpensive quartz clocks, which are not perfectly synchronized with the highly precise atomic clocks on the satellites. This introduces an unknown time offset (Δt), creating a distance bias $b_0 = c \cdot \Delta t$.
- **Pseudorange Equation**: Gr

$$\rho_i = \sqrt{(x_i - x)^2 + (y_i - y)^2 + (z_i - z)^2} + b_0 + \epsilon_i$$

- **Conclusion**: Because there are four unknown variables (x, y, z, b_0), a minimum of **four independent satellites** is required to formulate a system of four equations and accurately solve for the user's position.

What dilution of precision means geometrically:

- Geometrically, each satellite measurement acts as a spherical shell (with a thickness representing measurement uncertainty, UERE) centered on the satellite. The receiver is located at the intersection of these shells.
- **Low DOP (Good Geometry)**: The satellites are widely dispersed across the sky (e.g., one directly overhead and three evenly spaced near the horizon). The resulting intersection volume is compact and spherical, minimizing positional uncertainty.
- **High DOP (Poor Geometry)**: The satellites are clustered closely together in one area of the sky. The intersection volume elongates significantly along the line of sight, amplifying the range error into a much larger position error.

Task 2: Constellation Dynamics

Parameters:

- Earth's gravitational parameter: $\mu = 3.986 \times 10^{14} m^3/s^2$
- Galileo semi-major axis: $a = 29600\,km = 29.6 \times 10^6$

- Sidereal day: $T_{sidereal} \approx 23.9344\,hours = 86.164\,s$

1. **Compute Galileo Orbital Period (T):**

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} = 2\pi\sqrt{\frac{(29.6\times10^6)^3}{3.986\times10^{14}}} = 50681\,s = 14.08\,hours$$

2. **Compute Revolutions per Sidereal Day**

$$Revolutions = \frac{T_{sidereal}}{T} = \frac{23.9344\,h}{14.08\,h} = 1.7\,revs/sidereal\,day$$

3. **Comparison to GPS & Ground-Track Repeat Impact:**

- **Compared to GPS:** The GPS orbit $(a \approx 26560\,km)$ has a period of $T \approx 11.97\,h$, resulting in exactly **2.00 revolutions per sidereal day**.
- **Impact on Ground-Track Repeat:**
  - **GPS**: With a $2/1$ ratio, the GPS constellation repeats its exact ground track every **1 sidereal day**. A stationary receiver sees the exact same satellite geometry at the exact same time every day.
  - **Galileo**: The ratio is $1.7 = 17/10$, meaning a satellite completes 17 orbits every 10 sidereal days. Consequently, the ground track only repeats every **10 sidereal days**, providing greater spatial variation day-to-day.

Task 3: Relativity

1. **Theoretical Formulations**

- **Special Relativity (SR - Time Dilation due to velocity $v = \sqrt{\mu/a}$):**

$$\frac{\Delta f_{SR}}{f} = \frac{-v^2}{2c^2} = \frac{-\mu}{2ac^2}$$

- **General Relativity (GR - Gravitational Redshift due to potential difference between Earth's surface $r_e = 6378\,km$ and orbit a):**

$$\frac{\Delta f_{GR}}{f} = \frac{\Delta\Phi}{c^2} = \frac{\mu}{c^2}\left(\frac{1}{r_e} - \frac{1}{a}\right)$$

2. **Numerical Calculations $c = 2.9979\times10^8\,m/s$:**

- **SR Daily Shift:**

$$\frac{\Delta f_{SR}}{f} = \frac{-\mu}{2ac^2} = \frac{-3.986\times10^{14}}{2\times(29.6\times10^6)(2.9979\times10^8)^2} = -7.49\times10^{-11}$$

$$Daily\,SR\,drift = -7.49\times10^{-11}\times86400 = -6.4\,\mu s/day$$

- **GR Daily Shift:**

$$\frac{\Delta f_{GR}}{f}=\frac{\mu}{c^2}\left(\frac{1}{r_e}-\frac{1}{a}\right)=\frac{3.986\times10^{14}}{\left(2.9979\times10^{8}\right)^2}\left(\frac{1}{6.378\times10^{6}}-\frac{1}{29.6\times10^{6}}\right)=5.46\times10^{-10}$$

$$Daily\ GR\ drift=5.46\times10^{-10}\times86400=47.1\ \mu s/day$$

- **Net Relativistic Clock Drift:**

$$Net\ Drift=47.1-6.4=40.7\ \mu s/day$$

- **Uncorrected Range Error:**

$$Daily\ Range\ Error=c\cdot\Delta t=\left(2.9979\times10^{8}\ m/s\right)\times\left(40.8\times10^{-6}\ s\right)=12.2\ km/day$$

Task 4: DOP (Dilution of Precision)

1. Expected Position Error:

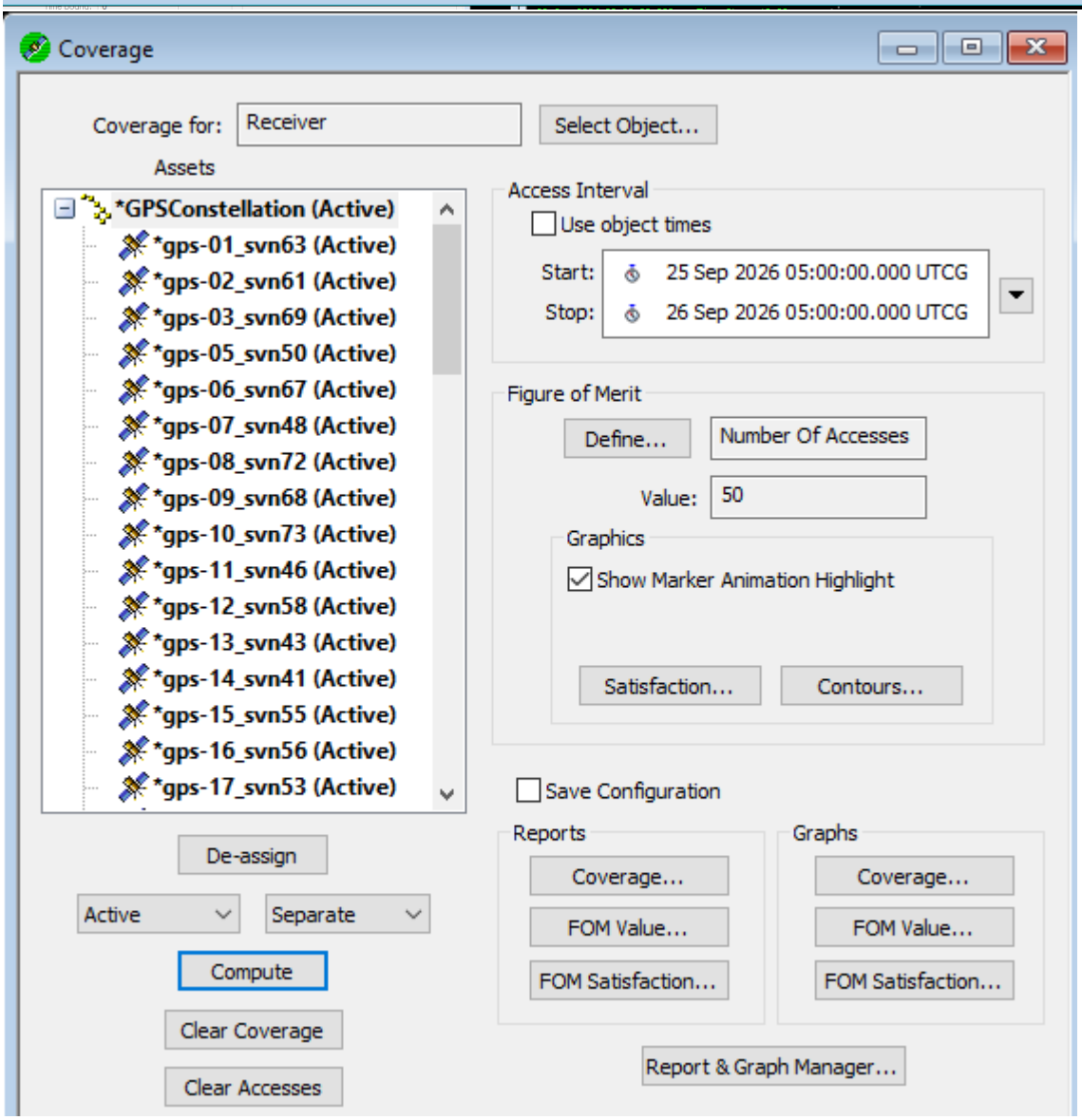
$$\sigma_{position}=PDOP\times UERE=2.3\times4=9.2\ m$$

2. Two Methods to Reduce DOP:

- **Increase the number of satellites in view**: Track additional satellites by utilizing a multi-constellation receiver (e.g., combining GPS, Galileo, GLONASS, and BeiDou).
- **Achieve a wider geometric spread across the sky**: Lower the elevation mask angle (e.g., from 15° down to 5°) to incorporate signals from satellites closer to the horizon, which spreads out the intersection volume.

Task 5: STK


Graph: Coverage for Receiver - All DOPs
Interval:
Scenario4 AnalysisInterval
Step: Using object's default time points
Show Step Value
Civil Air Patrol Use Only
DOP
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
6:00
9:00
12:00
15:00
18:00
21:00
26 Sat
3:00
Sep 25 Fri 2026
(UTCG)
GDOP - DOP Values for Receiver
PDOP - DOP Values for Receiver
HDOP - DOP Values for Receiver
VDOP - DOP Values for Receiver
TDOP - DOP Values for Receiver


Coverage
Coverage for:
Receiver
Select Object...
Assets
*GPSConstellation (Active)
*gps-01_svn63 (Active)
*gps-02_svn61 (Active)
*gps-03_svn69 (Active)
*gps-05_svn50 (Active)
*gps-06_svn67 (Active)
*gps-07_svn48 (Active)
*gps-08_svn72 (Active)
*gps-09_svn68 (Active)
*gps-10_svn73 (Active)
*gps-11_svn46 (Active)
*gps-12_svn58 (Active)
*gps-13_svn43 (Active)
*gps-14_svn41 (Active)
*gps-15_svn55 (Active)
*gps-16_svn56 (Active)
*gps-17_svn53 (Active)
Access Interval
Use object times
Start:
25 Sep 2026 05:00:00.000 UTCG
Stop:
26 Sep 2026 05:00:00.000 UTCG
Figure of Merit
Define...
Number Of Accesses
Value:
50
Graphics
Show Marker Animation Highlight
Satisfaction...
Contours...
Save Configuration
Reports
Coverage...
FOM Value...
FOM Satisfaction...
Graphs
Coverage...
FOM Value...
FOM Satisfaction...
Report & Graph Manager...
De-assign
Active
Separate
Compute
Clear Coverage
Clear Accesses